



Aror University of Art, Architecture, Design & Heritage Sukkur

BS Artificial Intelligence (Section-D)

Class: BS AI

Discrete Structures

Term: Spring 2026

Due Date: 13-04-2026

Assignment # 02

Max Marks: 10+5

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Date: 29-03-2026

Solve the following questions.

1.

Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express each of these sets with bit strings where the i th bit in the string is 1 if i is in the set and 0 otherwise.

- a) $\{3, 4, 5\}$
- b) $\{1, 3, 6, 10\}$
- c) $\{2, 3, 4, 7, 8, 9\}$

2.

Using the same universal set as in the last problem, find the set specified by each of these bit strings.

- a) 11 1100 1111
- b) 01 0111 1000
- c) 10 0000 0001

3.

What subsets of a finite universal set do these bit strings represent?

- a) the string with all zeros
- b) the string with all ones

4.

What is the bit string corresponding to the difference of two sets?

5.

What is the bit string corresponding to the symmetric difference of two sets?

6.

Show how bitwise operations on bit strings can be used to find these combinations of $A = \{a, b, c, d, e\}$, $B = \{b, c, d, g, p, t, v\}$, $C = \{c, e, i, o, u, x, y, z\}$, and $D = \{d, e, h, i, n, o, t, u, x, y\}$.

- a)** $A \cup B$ **b)** $A \cap B$
c) $(A \cup D) \cap (B \cup C)$ **d)** $A \cup B \cup C \cup D$

7.

How can the union and intersection of n sets that all are subsets of the universal set U be found using bit strings?

8. Let A and B be the multisets $\{(a, 3), (b, 2), (c, 1)\}$ and $\{(a, 2), (b, 3), (d, 4)\}$, respectively.

Find

- (a) $A \cup B$
(b) $A \cap B$
(c) $A \setminus B$
(d) $B \setminus A$
(e) $A \sqcup B$

9.

Suppose that A is the multiset that has as its elements the types of computer equipment needed by one department of a university and the multiplicities are the number of pieces of each type needed, and B is the analogous multiset for a second department of the university. For instance, A could be the multiset $\{107 \cdot \text{personal computers}, 44 \cdot \text{routers}, 6 \cdot \text{servers}\}$ and B could be the multiset $\{14 \cdot \text{personal computers}, 6 \cdot \text{routers}, 2 \cdot \text{mainframes}\}$.

- a)** What combination of A and B represents the equipment the university should buy assuming both departments use the same equipment?
b) What combination of A and B represents the equipment that will be used by both departments if both departments use the same equipment?
c) What combination of A and B represents the equipment that the second department uses, but the first department does not, if both departments use the same equipment?
d) What combination of A and B represents the equipment that the university should purchase if the departments do not share equipment?

10.

Fuzzy sets are used in artificial intelligence. Each element in the universal set U has a **degree of membership**, which is a real number between 0 and 1 (including 0 and 1), in a fuzzy set S . The fuzzy set S is denoted by listing the elements with their degrees of membership (elements with 0 degree of membership are not listed). For instance, we write $\{0.6 \text{ Alice}, 0.9 \text{ Brian}, 0.4 \text{ Fred}, 0.1 \text{ Oscar}, 0.5 \text{ Rita}\}$ for the set F (of famous people) to indicate that Alice has a 0.6 degree of membership in F , Brian has a 0.9 degree of membership in F , Fred has a 0.4 degree of membership in F , Oscar has a 0.1 degree of membership in F , and Rita has a 0.5 degree of membership in F (so that Brian is the most famous and Oscar is the least famous of these people). Also suppose that R is the set of rich people with $R = \{0.4 \text{ Alice}, 0.8 \text{ Brian}, 0.2 \text{ Fred}, 0.9 \text{ Oscar}, 0.7 \text{ Rita}\}$.

(a)

The **complement** of a fuzzy set S is the set $\bar{S}$, with the degree of the membership of an element in $\bar{S}$ equal to 1 minus the degree of membership of this element in S . Find $\bar{F}$ (the fuzzy set of people who are not famous) and $\bar{R}$ (the fuzzy set of people who are not rich).

(b)

The **union** of two fuzzy sets S and T is the fuzzy set $S \cup T$, where the degree of membership of an element in $S \cup T$ is the maximum of the degrees of membership of this element in S and in T . Find the fuzzy set $F \cup R$ of rich or famous people.

(c)

The **intersection** of two fuzzy sets S and T is the fuzzy set $S \cap T$, where the degree of membership of an element in $S \cap T$ is the minimum of the degrees of membership of this element in S and in T . Find the fuzzy set $F \cap R$ of rich and famous people.

11.

For each of the following functions, determine whether it is one-to-one and determine its range.

(a) $f: \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 2x + 1$

(b) $f: \mathbb{Q} \rightarrow \mathbb{Q}, f(x) = 2x + 1$

(c) $f: \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^3 - x$

(d) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$

(e) $f: [-\pi/2, \pi/2] \rightarrow \mathbb{R}, f(x) = \sin(x)$

(f) $f: [0, \pi] \rightarrow \mathbb{R}, f(x) = \sin(x)$

12.

Prove that for a finite set X any **one-to-one** function $f: X \rightarrow X$ is **bijective**.

13.

For each of the following functions $f: \mathbb{R} \rightarrow \mathbb{R}$, determine whether f is invertible, and, if so, determine f^{-1}

- (a) $f = \{(x, y) | 2x + 3y = 7\}$
- (b) $f = \{(x, y) | ax + by = c, b \neq 0\}$
- (c) $f = \{(x, y) | y = x^3\}$
- (d) $f = \{(x, y) | y = x^4 + x\}$

14.

Give an explicit formula for a function from the set of integers to the set of positive integers that is

- a)** one-to-one, but not onto.
- b)** onto, but not one-to-one.
- c)** one-to-one and onto.
- d)** neither one-to-one nor onto.

15.

Determine whether each of these functions is a bijection from $\mathbf{R}$ to $\mathbf{R}$.

- a)** $f(x) = -3x + 4$
- b)** $f(x) = -3x^2 + 7$
- c)** $f(x) = (x + 1)/(x + 2)$
- d)** $f(x) = x^5 + 1$

16.

Consider these functions from the set of students in a discrete mathematics class. Under what conditions is the function one-to-one if it assigns to a student his or her

- a)** mobile phone number.
- b)** student identification number.
- c)** final grade in the class.
- d)** home town.

17.

Let $S = \{-1, 0, 2, 4, 7\}$. Find $f(S)$ if

- a)** $f(x) = 1$.
- b)** $f(x) = 2x + 1$.
- c)** $f(x) = \lceil x/5 \rceil$.
- d)** $f(x) = \lfloor (x^2 + 1)/3 \rfloor$.

18.

Let $f(x) = \lfloor x^2/3 \rfloor$. Find $f(S)$ if

a) $S = \{-2, -1, 0, 1, 2, 3\}$.

b) $S = \{0, 1, 2, 3, 4, 5\}$.

c) $S = \{1, 5, 7, 11\}$.

d) $S = \{2, 6, 10, 14\}$.

19.

If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer.

If f and $f \circ g$ are onto, does it follow that g is onto? Justify your answer.

20.

Let $f(x) = ax + b$ and $g(x) = cx + d$, where a, b, c , and d are constants. Determine necessary and sufficient conditions on the constants a, b, c , and d so that $f \circ g = g \circ f$.

21.

Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ defined by

$f(x) = x^2$. Find

a) $f^{-1}(\{1\})$.

b) $f^{-1}(\{x \mid 0 < x < 1\})$.

c) $f^{-1}(\{x \mid x > 4\})$.

22.

Let $g(x) = \lfloor x \rfloor$. Find

a) $g^{-1}(\{0\})$.

b) $g^{-1}(\{-1, 0, 1\})$.

c) $g^{-1}(\{x \mid 0 < x < 1\})$.

23.

Let S be a subset of a universal set U . The **characteristic function** f_S of S is the function from U to the set $\{0, 1\}$ such that $f_S(x) = 1$ if x belongs to S and $f_S(x) = 0$ if x does not belong to S . Let A and B be sets. Show that for all $x \in U$,

a) $f_{A \cap B}(x) = f_A(x) \cdot f_B(x)$

b) $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)$

c) $f_{\bar{A}}(x) = 1 - f_A(x)$

d) $f_{A \oplus B}(x) = f_A(x) + f_B(x) - 2f_A(x)f_B(x)$

24.

Let $f(x) = 2x$ where the domain is the set of real numbers. What is

a) $f(\mathbf{Z})$?

b) $f(\mathbf{N})$?

c) $f(\mathbf{R})$?

25.

Find at least three different sequences beginning with the terms 1, 2, 4 whose terms are generated with a simple formula or rule. Assuming that your formula or rule is correct, determine the next 5 terms in each case.

26.

For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequences that begins with the given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence.

- (a) 3, 6, 11, 18, 27, 38, 51, 66, 83, 102, ...
- (b) 7, 11, 15, 19, 23, 27, 31, 35, 39, 43, ...
- (c) 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, ...
- (d) 0, 2, 8, 26, 80, 242, 728, 2186, 6560, 19682, ...
- (e) 1, 3, 15, 105, 945, 10395, 135135, 2027025, 34459425, ...
- (f) 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, ...
- (g) 2, 4, 16, 256, 65536, 4294967296, ...

27.

What are the values of these sums?

- (a) $\sum_{i=1}^5 (i + 1)$
- (b) $\sum_{i=0}^4 (-2)^i$
- (c) $\sum_{i \in \{1, 3, 5, 7\}} i^2$
- (d) $\sum_{i \in \{1, 3, 5, 7\}} \frac{1}{i}$

28.

What are the values of these sums?

- (a) $\sum_{i=1}^4 (i^2 + i)$
- (b) $\sum_{k=1}^4 k^2 + \sum_{k=1}^4 k$
- (c) $\sum_{i=0}^4 (3i^2 + 2i)$
- (d) $3 \sum_{k=0}^4 k^2 + 2 \sum_{k=0}^4 k$

29.

Find a formula for $\sum_{k=0}^m \lfloor \sqrt{k} \rfloor$ where m is a positive integer.

30.

What are the terms a_0, a_1, a_2 , and a_3 of the sequence $\{a_n\}$, where a_n equals

- a)** $2^n + 1?$ **b)** $(n + 1)^{n+1}?$
c) $\lfloor n/2 \rfloor?$ **d)** $\lfloor n/2 \rfloor + \lceil n/2 \rceil?$

31.

What are the terms a_0, a_1, a_2 , and a_3 of the sequence $\{a_n\}$, where a_n equals

- a)** $(-2)^n?$ **b)** $3?$
c) $7 + 4^n?$ **d)** $2^n + (-2)^n?$

32.

Find the first five terms of the sequence defined by each of these recurrence relations and initial conditions.

- a)** $a_n = 6a_{n-1}, a_0 = 2$
b) $a_n = a_{n-1}^2, a_1 = 2$
c) $a_n = a_{n-1} + 3a_{n-2}, a_0 = 1, a_1 = 2$
d) $a_n = na_{n-1} + n^2a_{n-2}, a_0 = 1, a_1 = 1$
e) $a_n = a_{n-1} + a_{n-3}, a_0 = 1, a_1 = 2, a_2 = 0$

33.

Compute each of these double sums.

- a)** $\sum_{i=1}^3 \sum_{j=1}^2 (i - j)$ **b)** $\sum_{i=0}^3 \sum_{j=0}^2 (3i + 2j)$
c) $\sum_{i=1}^3 \sum_{j=0}^2 j$ **d)** $\sum_{i=0}^2 \sum_{j=0}^3 i^2 j^3$

34.

Find $\sum_{k=100}^{200} k$.

Find $\sum_{k=99}^{200} k^3$.

35.

What are the values of the following products?

a) $\prod_{i=0}^{10} i$

b) $\prod_{i=5}^8 i$

c) $\prod_{i=1}^{100} (-1)^i$

d) $\prod_{i=1}^{10} 2$

36.

Find $\sum_{j=0}^4 j!$.

Find $\prod_{j=0}^4 j!$.